Anuranan Bharadwaj

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EDUCATION

Embry-Riddle Aeronautical University Daytona Beach, FL
Bachelor of Science in Aerospace Engineering - Jet Propulsion **May 2026**
Minor in Computer Science **GPA: 3.8**
Courses: Jet Propulsion| Structures| Materials| Aerodynamics| Thermodynamics| Stability & Control| OOP| DSA

EXPERIENCE

Turbomachinery Software Intern **January – April 2026**
Concepts NREC *White River Junction, VT*

- Designed and implemented **Python automation frameworks** using **AxCent's APIs** to execute automated testing of turbomachinery blade design workflows, replacing manual validation and generating structured **Excel-based verification reports.**
- Developed **automated validation pipelines** for **gas turbine performance** modeling tools, testing AxCent outputs across **diverse blade configurations** and accelerating **QA cycles** for proprietary turbomachinery design software.
- **Analyzing blade profile repositories** applied to 3D turbine rotor and stator models, validating key geometric parameters including **blade count, leading/trailing edge radius, gauge angle, inlet angle, and rotation direction (CW/CCW)** to ensure design consistency and reliability.

Undergraduate Research Assistant **May 2025 – April 2026**
Gas Turbine Lab – Embry-Riddle Aeronautical University *Daytona Beach, FL*

- Developed a **Python-Excel** tool to compute **turbine blade throat openings** and **deviation angles** via **Aungier's correlations** and **Mach-dependent thermodynamic cycle effects**, supporting gas turbine thermal performance analysis across operating conditions.
- Validated computational component analysis results against **experimental data**; automated **multi-stage turbine performance analysis** to accelerate **design iteration** and **aerodynamic characterization**.
- Designed and **CAD-modeled** a custom pitot rake for **multi-point total pressure measurement** in **turbine cascade rig**; supports field performance **data acquisition** and **instrumentation upgrades** for engine cycle validation.

Systems Engineering Intern **July – Aug 2024**
ABH Software *Assam, India*

- Engineered automated data reporting pipelines using **SQL** and **Java** to simulate real-time telemetry and logistics tracking; **reduced manual report processing by 30%** and **improved accuracy by 20%** using **Agile methodologies.**

PROJECTS

Aircraft Stability & Control Simulation | *MATLAB, Simulink, DATCOM, FlightGear*

- Built full MATLAB/Simulink flight simulation model with lookup tables for **aerodynamic derivatives** computed via USAF DATCOM; validated **longitudinal and lateral stability** under control inputs**.**
- Visualized flight **performance data** in FlightGear; confirmed effective pitch, roll, and yaw response.

Aircraft Wing Structural Analysis| *FEMAP, NX Nastran, Fusion 360*

- Performed FEA on full aircraft wing geometry (spars, ribs, stringers, skin) in FEMAP/NX Nastran under **aerodynamic load** conditions; **max displacement 0.15 in** with **Jacobian mesh quality > 0.6.**
- Interpreted **stress contours** and **displacement plots** to assess **structural margins** and **component stiffness**.

SKILLS & Development

Engineering Software: SolidWorks| CATIA V5/V6| FEMAP| CFTurbo| Simulink| MATLAB| Fusion 360| 3ds Max
Programming Language: Java| Python| HTML| CSS| JS| C#
Developer Tools: VS Code| Visual Studio| Git| PyCharm| Jupyter| Figma| Claude Code| N8N
Frameworks & Technologies: MERN| .NET| WPF| SQL| Pandas| Matplotlib| NumPy| Excel| JIRA| Confluence

- **Boeing Career Program** (BCP) Mentee – Class of 2025